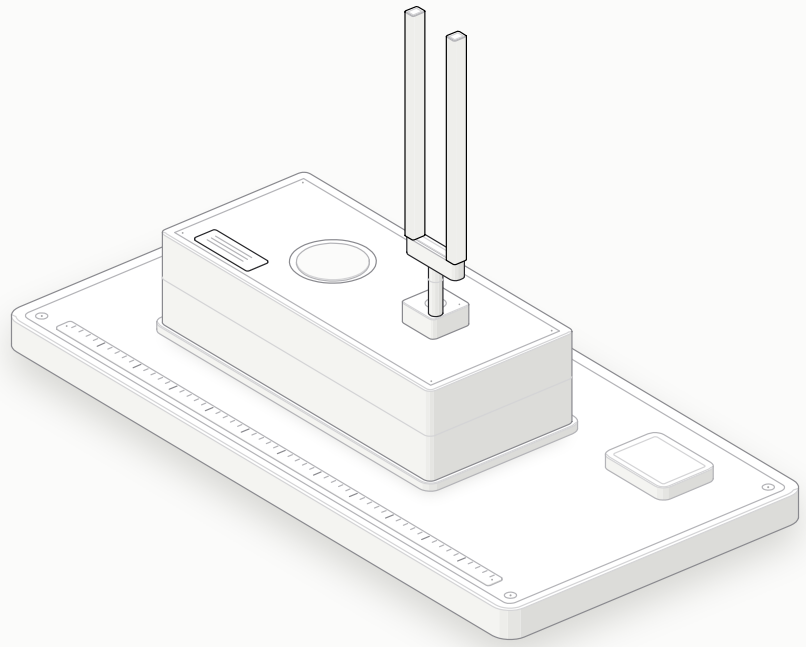
Fig 1 TUNING FORK ON ITS RESONANCE BOX



HOVER TO STRIKE · ARROWS: FORK 256 Hz · prongs 92 · 100%

- Prongs: swing in antiphase; shorter prongs, higher frequency ($f \propto 1/L^2$)
- Stem: moves up and down at the same frequency and drives the box

Strike the fork on the rubber pad and its prongs swing towards and away from each other. The stem moves up and down at the same frequency; pressed on a box or a sonometer, it drives that too. The 512 Hz fork's prongs are $1/\sqrt{2}$ as long as the 256 Hz fork's, because a prong's frequency goes as $1/L^2$. The sound dies away as the prongs lose energy.